

Max 2025
ZAK

Q1.

Ainee is the owner of several car shops and employs multiple managers who oversee the daily operations. She stores the data about her business in a relational database named "CARS." The database has three tables: SHOP, MANAGER, and CAR.

The SHOP table contains columns for ShopID, ManagerID, Address, Town, and TelephoneNumber.

The MANAGER table has columns for ManagerID, FirstName, LastName, DateOfBirth, and Wage.

The CAR table has columns for RegistrationNumber, Make, Model, NumberOfMiles, and ShopID.



a) Identify which fields in each table are primary keys and which ones are foreign keys.

<u>SHOP:</u>	<u>MANAGER:</u>	<u>CAR:</u>
PK: ShopID	PK: ManagerID	PK: RegistrationNumber
FK: ManagerID	FK: —	FK: ShopID

[2]

b) Explain how access rights can be used to safeguard the data in Ainee's database against unauthorized access.

Access rights are used to assign specific permissions to different users based on their roles.

For example, managers are given permission to access only the data related to their own shop, ensuring they cannot view or modify other shops data.

Ainee as the owner is given full administrative access to all data in the database.

Additionally, authentication measures such as secure passwords are implemented to ensure that only authorised users can access the database system.

[3]

c) (i) Ainee uses DDL and DML statements in her database. Complete the following DML statements:

To retrieve the number of cars for sale in each shop.

```

SELECT ShopID, COUNT(*) AS NumberOfCars
FROM CAR
GROUP BY ShopID;
  
```

[3]

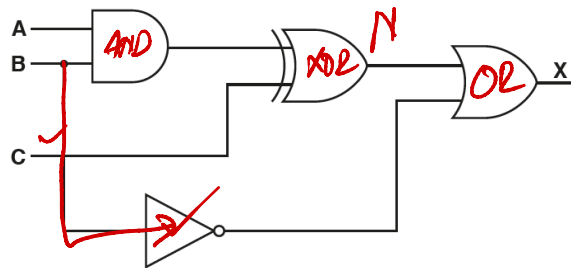
(ii) To insert a new record for a car with the following data: RegistrationNumber: 123AA, Make: Tiger, Model: Lioness, NumberOfMiles: 10500, ShopID: 12BSTREET.

INSERT INTO CAR (RegistrationNumber, Make, Model, NumberOfMiles, ShopID)
VALUES ('123AA', 'Tiger', 'Lioness', 10500, '12BSTREET');

[2]

Q2.

A logic circuit is shown:



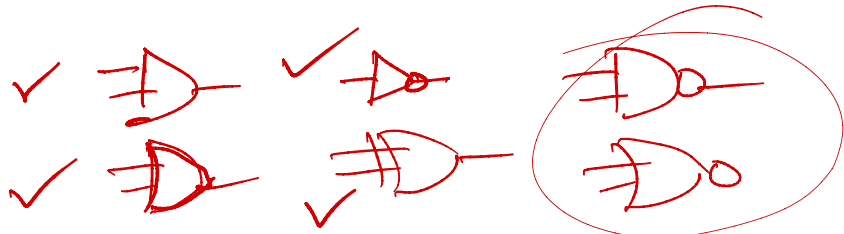
(a) Write the logic expression for the logic circuit.

$$X = ((A \text{ AND } B) \text{ XOR } C) \text{ OR NOT } B$$

[3]

(b) Complete the truth table for the given logic circuit.

A	B	C	AB	Working space		X
				N	B'	
0	0	0	0	0	1	1
0	0	1	0	1	1	1
0	1	0	0	0	0	0
0	1	1	0	1	0	1
1	0	0	0	0	1	1
1	0	1	0	1	1	1
1	1	0	1	1	0	1
1	1	1	1	0	0	0



[2]

c) Identify one logic gate not used in the given logic circuit. Draw the symbol for the logic gate and complete its truth table.

Logic gate: NAND

Symbol: 

Truth table:

A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0

[3]

Q3.

The following table shows part of the instruction set for a processor that has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		Explanation
Op code	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC.
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC.
LDI	<address>	Indirect addressing. The address to be used is at the given address. Load the contents of this second address to ACC.
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the Index Register. Copy the contents of this calculated address to ACC.
LDR	#n	Immediate addressing. Load the number n to IX.
STO	<address>	Store the contents of ACC at the given address.
STX	<address>	Indexed addressing. Form the address from <address> + the contents of the Index Register. Copy the contents of ACC to this calculated address.
ADD	<address>	Add the contents of the given address to ACC.
INC	<register>	Add 1 to the contents of the register (ACC or IX).
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX).
JMP	<address>	Jump to the given address.
CMP	<address>	Compare the contents of ACC with the contents of <address>.
CMP	#n	Compare the contents of ACC with number n.
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True.
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False.
IN		Key in a character and store its ASCII value in ACC.
OUT		Output to the screen the character whose ASCII value is stored in ACC.
END		Return control to the operating system.

Study the following low level language code and answer subsequent questions related to the same. These instructions are using three arrays (Array1, Array2 & Array3), which are saved in memory locations 100, 200 and 300 respectively.

Handwritten notes above the code table:

- 100 3 → 200 1 ✓ 300 4 ✓
- 101 4 201 2 301 6
- 102 5 202 8 302 13

	LDR #0 ✓	; load initial value to IX
	LDM #300 ✓	; load the address of first element of Array3 into C
	STO C ✓	
	LDM #0	
	STO D	; initiate loop counter
LOOP:	LDX 100	; load content of next element of Array1 into A
	STO A	
	LDX 200	; load content of next element of Array2 into B
	STO B	
	LDD A	; load content of A into accumulator
	<i>SUB B</i> ← ADD B	; add content of B to accumulator
	STX C	; store the result from the indexed memory address stored in C
	LDD D	; loop count
	INC ACC	
	CMP #3	; check if all elements of arrays 1 & 2 are read and added to 3rd
	JPE ENDD	; end if at end of array
	STO D	
	INC IX	; increment pointer to next elements in Array1 & Array2
	JMP LOOP	; repeat loop if not at the end of array
ENDD:	END	; end of program
100	3	
101	4	
102	5	
200	1 ✓	
201	2	
202	8	

(a) Dry-run this assembly language program using the trace table.

[illegible]

(b)

(i) What is the purpose of the given assembly language code?

.....
The program performs element wise addition of two arrays that is array1 and array2 and stores the result which is sums into the third array.
.....
.....
.....
.....
.....
.....
.....

[4]

(ii) What is the size of the three arrays?

.....
3 elements. per array.
.....

[1]

(iii) What is the significance of the label "LOOP:" in the code?

.....
The label LOOP marks the beginning of the loop that repeats the addition process for each pair of elements in the arrays.
.....

[1]

(iv) What does the instruction "CMP #3" do?

.....
It compares the value in accumulator with 3 to determine if all array elements have been processed.
.....

[1]

(v) Which instruction would you modify, and to what instruction, to subtract the content of location B from A and store the result in address saved in C?

.....
[1]

Q4:

(a) Give denary equivalents of the following IEC binary prefixes used to represent quantities of digital information:

(i) 1 Kibi: 2^{10} 1024 Bytes

[1]

(ii) 1 Mebi: 2^{20} 1048576 Bytes.

[1]

(iii) 1 Gibi: 2^{30} 1073741824 Bytes.

[1]

(b) A computer's hard disk has a capacity of $\frac{1}{2}$ terabytes (TB). Convert this capacity to the nearest number of gibibytes (GiB), using the appropriate IEC binary prefix.

$$\frac{1}{2} \text{ TB} = 500 \text{ GB}$$

$$1 \text{ GB} = 2^{30} = 1073741824$$

$$1 \text{ GiB} = 2^{30} \text{ B} = 1073741824 \text{ B}$$

$$1 \text{ TB} = 10^9 \text{ B} = 1000,000,000 \text{ B}$$

$$500 \text{ GB} = \frac{500 \times 10^9}{2^{30}}$$

$$\Rightarrow 465.66 \text{ GiB} \quad [2]$$

(c) Perform the following binary addition using given bits only. Show your working.

(i) 11010000 + 10110000

$$\begin{array}{r} 111 \\ 11010000 \\ + 10110000 \\ \hline 100000000 \\ \text{OF} \end{array}$$

-128

[2]

(ii) Determine whether an overflow occurred during the addition. Explain your reasoning briefly.

9th bit overflow

b/c result requires 9th bit,
that is more than 8 bits.

[1]

(iii) Convert the result to denary.

$$\begin{array}{r} \text{128 64 32 16 8 4 2 1} \\ (100000000)_2 = (-128)_{10} \end{array}$$

2's

[1]

(iv) Convert the result to hexadecimal.

$$\begin{array}{r} 1000 \quad 0000 \\ \hline 8 \quad 0 \end{array}$$

[1]

Q5:

ZAK wants to protect the sensitive data on his company's computer network from unauthorized access and malware attacks.

(a) Explain the importance of data encryption and access controls in ensuring data security on a computer network.

Explanation:

Data encryption ensures that sensitive information is transformed into a coded form that can only be read by authorised users processing the correct description key.

Access controls restrict access to data based on user rules, allowing only authorised individuals to view or modify sensitive information.

Together, they enhance security by preventing unauthorised access and reducing the risk of data breaches or information theft.

[3]

(b) ZAK is considering implementing antivirus software on his company's computers. Describe how antivirus software works to protect against malware attacks and provide an example of a common type of malware.

Explanation:
Antivirus software protects a computer by scanning files and programs for known malware signatures or suspicious behaviour patterns.

.....
Quarantine or deletes detected malware, provides real time protection against threats, and updates its database regularly to recognise new types of attacks.

Example:
Example of malware is spyware. It is a malicious software that secretly collects user information without consent. [3]

(c) Two common types of cyber-attacks are phishing and pharming. Describe the difference between phishing and pharming.

Phishing: it is when a user is tricked into revealing sensitive information by fake emails or websites. For example, an email pretending to be from a bank asking for login details.

.....
Pharming: it is when a user is redirected to a fake website even after entering the correct web address. For example a fake banking website that captures login credentials.

.....
[2]

Q6.

Harim is starting her first job as a software developer for a multinational company.

(a) Harim has been advised to join a professional ethical body.

Explain the benefits to Harim of joining a professional ethical body.

Joining a professional ethical body demonstrate Harim's commitment to maintaining high professional and ethical standards.
It enhances her reputation with employers and colleagues, provides resources and support for handling ethical issues, and offers networking and professional development opportunities through conferences, workshops, and certifications.

[3]

(b) Harim is shown the software she will be working on. She is unfamiliar with the Integrated Development Environment (IDE) she is required to use.

(i) Describe the ethical ways in which Harim can act in this situation.

Hareem should act ethically by asking for guidance from her colleagues or supervisors, studying the IDE through available resources, seeking training if necessary, and being transparent about her inexperience to ensure she can contribute effectively to the team.

[2]

(ii) In addition to debugging tools, identify three other tools or features commonly found in an IDE that support the writing of a program.

- 1 Syntax highlighting
- 2 Auto complete
- 3 Code folding, expand and collapse sections of code.

[3]

(c) Harim is part of a team writing a program. She finds an error in part of the program that has already been tested. She decides not to tell anyone because she is worried about the consequences.

Explain the reasons why Harim acted unethically in this situation.

Haim acted unethically because by not reporting the error, she jeopardise the quality, reliability, and security of the software.

Her actions could harm users, damage the companies reputation, and violate professional standards of honesty, accountability and teamwork.

[2]

(d) Harim's team use language translators.

Complete the descriptions of language translators by writing the missing words.

Compilers are usually used when a high-level language program is complete. They translate all the code at the same time and then run the program.

They produce executable(.exe) files that can be run without the source code.

Interpreters translate one line of a high-level language program at a time, and then run that line of code. They are most useful while developing the programs because errors can be corrected and then the program continues from that line.

Assemblers are used to translate assembly code into machine code.

[4]

Blank Page

Q7.

(a) Table below shows four stages in converting sound into a digital form.

Show the correct order for the stages by labelling them with the numbers 1 – 4

(1 being the first stage).

Stage	Order (1-4)
Binary representation of level stored	4
microphone picks up sound waves	1
value read at specific point and rounded to a level	3
converted to an electrical analogue signal	2

[3]

(b) Describe how a black and white image could be represented as a bitmap in binary.

A black-and-white image is represented as a bit map by dividing the image into a grade of pixels.

Each pixel is assigned a binary value, that is zero for black and one for white.

The binary values are stored sequently along with information about the image dimensions and file structure.

[3]

(c) A screen contains black text on a solid white background. One line of the screen might be represented as follows with W representing white pixel and B representing a black pixel.

WWWWWWBBBWWWWWWWWWWWWBBBWWBBBWWWW

A compression technique is applied to the line of data above and results in the following output:

6W3B12W6B1W3B4W

State what data compression algorithm has been applied.

RLE

[1]

(d) Another data compression technique is JPEG, which is a lossy compression algorithm.

Explain what is meant by lossy compression.

Lossy compression reduces the file size by permanently removing some data that is less important.
~~The decompressed file is not identical to the original, resulting in a slight loss of quality.~~
 It is commonly used for compressing media files such as images or audio.

[1]

Q8.

Fill in the blanks.

The two main types of networks are LAN and WAN. The key difference between them is that first one is confined to a small geographic area, while the other can cover a larger geographic area, such as a city, country, or even the entire world.

The two main models of networked computers are Client-Server and peer-to-peer. In the first model, one computer (the server) provides services or resources to other computers (the clients) on the network. In the other, all computers on the network are equal and can share resources with each other.

The different types of subnetwork models include bus, star, mesh, and hybrid topologies. Each has its own benefits and drawbacks, such as scalability, reliability, and ease of implementation.

The hardware used in LAN networks includes switches, servers, NICs, WNICs, WAPs, cables, bridges, and repeaters. These devices are used to connect computers and other devices on the network, and facilitate communication between them.

The role of a router in a network is to direct data packets between different networks. Ethernet is a set of rules and protocols that govern how data is transmitted over a LAN network, and includes mechanisms to avoid collisions, such as

CSMA/CD

Cloud Computing is a model for delivering on-demand computing resources over the internet. Public Cloud are available to anyone over the internet, while private Cloud are reserved for a single organization or business. The benefits include scalability, flexibility, and cost-effectiveness, while drawbacks include security and privacy concerns.

[4]